EE2021: Assignment -2 (Hardcopy submission by Sept 30th, 2025)

1. List two reasons why desert sand cannot be used for making pure Silicon wafers (which is further processed through 1000s of steps to make semiconductor products like ICs). Your answer should NOT contain more than 3 sentences

2. The Laplacian and angular momentum operator for spherical coordinates is given below

$$L_z = -i\hbar\frac{\partial}{\partial\phi}$$



$$\nabla^2 \equiv \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{r^2\sin^2\phi}\frac{\partial^2}{\partial\theta^2}+\frac{1}{r^2\sin\phi}\frac{\partial}{\partial\phi}\left(\sin\phi\frac{\partial}{\partial\phi}\right).$$

   Simplify the Laplacian by assuming the width and thickness of the ring is very small compared to its radius. Write the Schrodinger equation for a particle in a ring of circumference, L. Consider the particle is free to move only around the ring, find its wavefunction, angular momentum & energy eigenvalues and probability current density.

   How are the answers different compared to the 1-D free particle problem solved in class?

3. There are situations in quantum mechanics where the mass (relationship between the applied force and acceleration felt) of a charged particle is not constant across space. This occurs specifically in the analysis of semiconductor heterostructures where the **effective mass** (Wait for a couple of classes to understand what on earth is that) is different in different materials. For the case where mass m(z) varies with z, we can postulate the 1-D Hamiltonian as

$$\hat{H} = -\frac{\hbar^2}{2}\frac{d}{dz}\left(\frac{1}{m(z)}\frac{d}{dz}\right)+V(z)$$

   a) Derive the expression for particle/probability current density using the above Hamiltonian.
   b) What boundary conditions should be used for the wavefunction at a step/discontinuity in potential energy (which happens at material interfaces) to ensure the continuity of particle/probability current density.